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B.Tech. / M.Tech. (Integrated) DEGREE EXAMINATION, NOVEMBER 2024

Third Semester

21MAB201T - TRANSFORMS AND BOUNDARY VALUE PROBLEMS

(For the candidates admitted from the academic year 2021-2022 to 2023 - 2024)

Note:

(i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.

(ii) **Part - B & Part - C** should be answered in answer booklet.

Time: 3 hours

Max. Marks: 75

PART - A (20 x 1 = 20 Marks)

Answer ALL Questions

Marks BL CO PO

- The partial differential equation formed by eliminating arbitrary constants in $z = ax + by$ is 1 1 1 1
 A) $z = px + qy$ B) $z = qx + py$
 C) $z = px - qy$ D) $z = qx - py$
- The partial differential equation formed by eliminating arbitrary function in $z = f(xy)$ is 1 2 1 1
 A) $xp - yq = 1$ B) $xp = yq$
 C) $p + q = 1$ D) $p - q = 0$
- The solution which has number of arbitrary constants equal to number of independent variables is called 1 1 1 1
 A) general integral B) complete integral
 C) particular integral D) singular integral
- The complete integral of $q = 2py$ is 1 2 1 1
 A) $z = ax + ay^2 + b$ B) $z = ax^2 - ay^2 + b$
 C) $z = ax + by$ D) $z = 2xy$
- The root mean square value of $f(x) = x$ in $(0, L)$ is 1 2 2 1
 A) $\frac{L^2}{\sqrt{3}}$ B) $\frac{L}{3}$
 C) 1 D) $\frac{L}{\sqrt{3}}$
- The value of the Fourier series of $f(x) = x$ in $0 < x < 2l$ at $x = 0$ is 1 2 2 1
 A) 0 B) l
 C) 1 D) $\frac{l}{2}$
- The Fourier coefficient a_0 in the Fourier series expansion of $f(x) = (\pi - x)$ in $(0, 2\pi)$ is 1 2 2 1
 A) $\frac{1}{\pi}$ B) 1
 C) 0 D) $\frac{\pi}{2}$
- In the Fourier series Expansion of $f(x) = x^2$, $-\pi < x < \pi$, the value of b_n is 1 1 2 1
 A) 0 B) 1
 C) $\frac{1}{n}$ D) $\frac{1}{n^2}$

9. In the partial differential equation $\frac{\partial^2 y}{\partial t^2} = a^2 \frac{\partial^2 y}{\partial x^2}$, a^2 stands for
 A) $\frac{T}{m}$ B) $\frac{\kappa}{c}$
 C) $\frac{m}{T}$ D) kc 1 1 3 1
10. How many initial and boundary conditions are required to solve $\frac{\partial u}{\partial t} = \alpha^2 \frac{\partial^2 u}{\partial x^2}$
 A) Four B) Two
 C) Three D) Five 1 1 3 1
11. In one dimensional wave equation the tension T caused by stretching the string before fixing it at the end points is assumed to be
 A) Large B) Decreasing
 C) Constant D) Zero 1 1 3 1
12. The partial differential equation $\frac{\partial^2 z}{\partial x^2} - 5 \frac{\partial^2 z}{\partial y^2} = 0$ is classified as
 A) Elliptic B) parabolic
 C) Hyperbolic D) Circle 1 1 3 1
13. If $F(f(x)) = F(s)$, then $F[f(x) \cos ax] =$
 A) $\frac{1}{2} [F(s+a) * F(s-a)]$ B) $\frac{1}{2} [F(s+a) - F(s-a)]$
 C) $\frac{1}{2} [F(s+a) + F(s-a)]$ D) $\frac{1}{2} [F(sa) + F(s+a)]$ 1 1 4 1
14. If $F(f(x)) = F(s)$, then $F(x^n f(x)) =$
 A) $(-i)^n \frac{d^n}{ds^n} F(s)$ B) $(i)^n \frac{d^n}{ds^n} F(s)$
 C) $(-i) \frac{d^n}{ds^n} F(s)$ D) $\frac{d^n}{ds^n} F(s)$ 1 1 4 1
15. The Fourier cosine transform of e^{-ax} , $a > 0$ is
 A) $\sqrt{\frac{1}{\pi}} \frac{a}{a^2+s^2}$ B) $\sqrt{\frac{2}{\pi}} \frac{a}{a^2+s^2}$
 C) $\sqrt{\frac{1}{\pi}} \frac{s}{a^2+s^2}$ D) $\sqrt{\frac{2}{\pi}} \frac{s}{a^2+s^2}$ 1 2 4 1
16. If $F_s(s)$, $G_s(s)$ are the Fourier sine transforms of $f(x)$ and $g(x)$ respectively, then $\int_0^\infty f(x) g(x) dx =$
 A) $\int_0^\infty F_s(s) G_s(s) ds$ B) $\int_{-\infty}^\infty F_s(s) G_s(s) ds$
 C) $\sqrt{\frac{2}{\pi}} \int_0^\infty F_s(s) G_s(s) ds$ D) $\sqrt{\frac{2}{\pi}} \int_{-\infty}^\infty F_s(s) G_s(s) ds$ 1 1 4 1
17. The value of $Z[(-2)^n]$ is
 A) $\frac{z}{z+2}$ B) $\frac{z}{z-2}$
 C) $\frac{z^2}{z+2}$ D) $\frac{2}{z+2}$ 1 1 5 1
18. $Z[f(n) * g(n)] =$
 A) $F(z)G^{-1}(z)$ B) $F^{-1}(z)G^{-1}(z)$
 C) $F(z).G(z)$ D) $F^{-1}(z).G(z)$ 1 1 5 1
19. The value of $Z^{-1}\left[\frac{z}{(z-1)^2}\right]$ is
 A) $n+1$ B) $n-1$
 C) n D) $\frac{1}{n}$ 1 1 5 1
20. The Poles of $f(z) = \frac{z^n}{(z-1)(z-2)}$ are
 A) $z=1, 0$ B) $z=0, 2$
 C) $z=-1, -2$ D) $z=1, 2$ 1 1 5 1

PART - B (5 x 8 = 40 Marks)
Answer ALL Questions

Marks BL CO PO

- 21 a. Solve $x(y-z)p + y(z-x)q = z(x-y)$. 8 3 1 2
(OR)
b. Solve $(D^2 - 2D \cdot D')z = e^{x+2y} + \sin(2x+3y)$. 8 3 1 2
- 22 a. Find the Fourier series for the function $f(x) = x^2$, $-\pi < x < \pi$. Hence find the value of $\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots$. 8 3 2 2
(OR)
b. Find the Half-range sine series for the function $f(x) = x(\pi-x)$ in the interval $(0, \pi)$ and hence deduce the value of $\frac{1}{1^3} - \frac{1}{3^3} + \frac{1}{5^3} - \frac{1}{7^3} + \dots$. 8 3 2 2
- 23 a. A tightly stretched string with fixed ends $x=0$ and $x=l$ is initially in a position $y(x,0) = \sin^3\left(\frac{\pi x}{l}\right)$. It is released from rest from this position, find the displacement of the string $y(x,t)$. 8 4 3 2
(OR)
b. Heat flows through a uniform rod of length l which has its sides insulated and the temperature at the ends kept at zero. If the initial temperature at the interior points of the rod is given by $k(l-x)$, find the temperature distribution in the rod at time t . 8 4 3 2
- 24 a. Find the Fourier Transforms of the function $f(x) = \begin{cases} 1 & \text{for } |x| < 1 \\ 0 & \text{for } |x| > 1 \end{cases}$ and hence evaluate $\int_0^\infty \frac{\sin x}{x} dx$. 8 3 4 2
(OR)
b. Evaluate $\int_0^\infty \frac{x^2}{(x^2+a^2)(x^2+b^2)} dx$, ($a, b > 0$) using transforms techniques. 8 3 4 2
- 25 a. Find (i) $Z(n^2)$, (ii) $Z(\cos n\theta)$. 8 2 5 2
(OR)
b. Find $Z^{-1}\left[\frac{z^2}{(z-a)(z-b)}\right]$, $a > b$ using Convolution Theorem. 8 2 5 2

PART - C (1 x 15 = 15 Marks)
Answer ANY ONE Question

Marks BL CO PO

26. Compute the first three harmonics of the Fourier series for the following data: 15 4 1 2
- | | | | | | | |
|-----|-----|-----------------|------------------|-------|------------------|------------------|
| x | 0 | $\frac{\pi}{3}$ | $\frac{2\pi}{3}$ | π | $\frac{4\pi}{3}$ | $\frac{5\pi}{3}$ |
| y | 1.0 | 1.4 | 1.9 | 1.7 | 1.5 | 1.2 |
27. (i) Solve the difference equation $y_{n+2} - 7y_{n+1} + 12y_n = 2^n$ given $y_0 = y_1 = 0$, using z-transforms. (8 Marks) 15 4 5 2
(ii) Find the inverse Z-transform of $\frac{z}{(z+1)^2}$ by long division method. (7 Marks)
